IA4

(1) A free space-silver interface has $E_0^i = 100$ V/m on the free-space side. The frequency is 15 MHz and the silver constants are $\epsilon_r = \mu_r = 1$ and $\sigma = 61.7$ MS/m. Determine E_0^r and E_0^t at the interface.

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M1
$\sigma = 0$
$\epsilon_1 = \epsilon_0$
$\mu_1 = \mu_0$

M2
$f = 15\ MHz = 15 \times 10^6\ Hz$
$\sigma_2 = 61.7\ MS/m$
$\epsilon_2 = \epsilon_r \epsilon_0 = 1 \cdot \epsilon_0 = \epsilon_0$
$\mu_2 = \mu_r \epsilon_0 = 1 \mu_0 = \mu_0$

$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377$
$E_0^i = 100$ V/m
$\omega = 2\pi f = 94.25 \times 10^7$ rad/s

Loss tangent
$\tan\delta = \frac{\sigma_2}{\omega \epsilon_2} = 7.4 \times 10^{10} >> 1$

Silver → excellent conductor

$\eta_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2}}$

$\eta_2 = 9.797 \times 10^{-4} + j\, 9.797 \times 10^{-4}\ \Omega$

Reflection coeff $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \approx \frac{-\eta_1}{\eta_1} = -1$
$= 1 \angle 180^\circ$

Transmission coeff $\tau = \frac{2\eta_2}{\eta_1 + \eta_2} \approx \frac{2\eta_2}{\eta_1} = 5.197 \times 10^{-6} + j\, 5.197 \times 10^{-6}$

$E_0^r = \Gamma E_0^i = (-1)(100) = -100$ V/m

$E_0^+ = \tau(100) = (5.2 + 5.2j) \times 10^{-4}$ V/m

$|E_0^+| = 7.35 \times 10^{-4}$ V/m

(2) Find the amplitude of displacement current density in metallic conductor at 60Hz if $\varepsilon = \varepsilon_0$, $\mu = \mu_0$, $\sigma = 5.8 \times 10^7$ S/m and $J = \sin(377t - 117.1z)\ \mathbf{a}_x$ mA/m^2.

$f = 60$ Hz, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$

$\sigma = 5.8 \times 10^7$ S/m

$J = \sin(3.77t - 117.1z)\ a_x$ mA/m^2

$\omega = 377 = 2\pi \times 60$ rad/s

$$|J_c| = 1m\ A/m^2 \Rightarrow 10^{-3}\ A/m^2$$

$$|J_d| = |J_c| \times \frac{\omega\varepsilon}{\sigma}$$

$$|J_d| = 10^{-3} \times 5.75 \times 10^{-17}$$

$$|J_d| = 5.75 \times 10^{-20}$$

3

In phasor form the electric filed intensity of a uniform plane wave in free space is expressed as $E = (40 - j30)e^{-j20z}\ a_x$ V/m. Find

a) Angular frequency ω
b) Phase constant β
c) Frequency f
d) Wavelength λ
e) Magnetic field intensity H

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$$E = (40 - j30)\, e^{-j20z}\, a_x\ V/m$$

$$\beta = 20\ rad/m$$

(a) $\omega = \beta \times C = 6 \times 10^9\ rad/s$

(b) $\beta = 20\ rad/m$

(c) $f = \frac{\omega}{2\pi} = 9.55 \times 10^8\ Hz$

$= 955\ MHz$

(d) $\lambda = \frac{2\pi}{\beta} = 0.314\ m$

(e) $H_0 = \frac{E_0}{\eta} = \frac{40 - 30j}{377}$

$$H = (0.106 - j0.0796)\, e^{-j20z}\, a_y\ A/m$$

4

A 1 MHz plane wave is propagating in fresh water having $\mu_r = 1$, $\epsilon_r = 81$ and $\epsilon'' = 0$ in $\mathbf{a}_z$ direction. Determine (i) phase constant β, (ii) wavelength, (iii) velocity, (iv) impedance of the medium, and (v) expressions of electric and magnetic fields. Assume the amplitude $E_{x0} = 1$.

$f = 1\ MHz$, $\mu_r = 1$, $\varepsilon_r = 81$, $\varepsilon'' = 0$ → lossless

$E_{x0} = 1$

$$V = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0 81 \varepsilon_0}} = \frac{1}{9\sqrt{\mu_0\varepsilon_0}} = \frac{3\times10^8}{9}$$

$$= 3.33 \times 10^7\ m/s$$

$$\omega = 2\pi f = 2\pi \times 10^6\ rad/s$$

(i) $\beta = \frac{\omega}{V} = \frac{2\pi \times 10^6}{1/3\times10^8} = 0.06\pi\ rad/m \approx 0.1885\ rad/m$

(ii) $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.06\pi} = 33.33\ m$

(iii) $V = 3.33 \times 10^7$

(iv) $\eta = \sqrt{\mu/\varepsilon} = \frac{1}{9}\sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{\eta_0}{9} = \frac{377}{9}$

(iii) $v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = \frac{1}{\sqrt{4\pi\times10^{-7}\times\frac{10^{-9}}{36\pi}}} = 3\times10^8$ m/s

(iv) $\eta = \sqrt{\mu/\varepsilon} = \frac{1}{9}\sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{\eta_0}{9} = \frac{377}{9}$

$\approx 41.89\,\Omega$

(v) $E(z,t) = a_x E_{x0}\cos(\omega t - \beta z)$

$E(z,t) = \cos(2\pi\times10^6 t - 0.1885z)\,a_x$ V/m

$H(z,t) = \frac{1}{41.89}\cos(2\pi\times10^6 t - 0.1885z)\,a_y$ A/m

$= 0.02387$ A/m $\cos(2\pi\times10^6 t - 0.1885z)\,a_y$ A/m

5

Given a non-magnetic material having $\varepsilon_r' = 3.2$ and $\sigma = 1.5\times10^{-4}$ S/m. Find the values at 3 MHz for (i) loss tangent (ii) attenuation constant (iii) phase constant (iv) Intrinsic impedance.

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$f = 3$ MHz $\quad \varepsilon_r' = 3.2 \quad \sigma = 1.5\times10^{-4}$ S/m

Non magnetic $\mu = \mu_0$

i) $\tan\delta = \frac{\sigma}{\omega\varepsilon'} = \frac{1.5\times10^{-4}}{2\pi\times3\times10^6\times\varepsilon_0\times3.2} = 0.281$

ii) $\alpha = \omega\sqrt{\frac{\mu_0\varepsilon_0\varepsilon_r}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]}$

(iii) $\beta = \omega\sqrt{\frac{\mu_0\varepsilon_0\varepsilon_r}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]}$

$\alpha = 0.0154$ Np/m

$\beta = 0.113$ rad/m

(iv) $\eta = \sqrt{\frac{\mu}{\varepsilon_r}} = \frac{377}{\sqrt{3.2}} = 210.75\,\Omega$

$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon'}} = \sqrt{\frac{\mu}{\varepsilon'}}\cdot\frac{1}{\sqrt{1-j(\tan\delta)}}$

$\frac{1}{\sqrt{1-j\tan\delta}}$ $\qquad 1 - j0.281 = 1.0387\angle -15.69$

$r = 1.0387, \quad \theta = -15.69$

$\sqrt{1-j0.281} \approx 1.0197\angle -7.845$

$\frac{1}{1.0197\angle -7.845} \Rightarrow 0.9817\angle 7.845$

$\eta = (210.75)(0.9817\angle 7.845)$

$\eta \approx (204.8 + j28.2)\,\Omega$